

Pattern Recognition Project

Comparative Study of Parametric and Non-Parametric Classifiers on the Two Moons Dataset

Dataset:

- Use Two moons dataset to accomplish the following

```
from sklearn.datasets import make_moons  
X, y = make_moons(n_samples=500, noise=0.25, random_state=42)
```

Task 1:

- Estimate distribution parameters for the dataset using MLE
- Classify each class using MAP classifier.
- Plot decision boundary for the classifier.

Task 2:

- Estimate distribution parameters for the dataset using MAP parameter estimation
- Classify each class using MAP classifier.
- Plot decision boundary for the classifier.

Task 3:

- Estimate distribution parameters for the dataset using EM Algorithm
- Classify each class using MAP classifier.
- Plot decision boundary for the classifier.

Task 4:

- Estimate distribution parameters for the dataset using Parzen-window (KDE)
- Classify each class using MAP classifier.
- Plot decision boundary for the classifier.

Task 5:

- Estimate distribution parameters for the dataset using KNN
- Classify each class using MAP classifier.
- Plot decision boundary for the classifier.

Task 6: Comparative Analysis

- Compare the results from all tasks (MLE, MAP, EM, KDE, KNN).
- Discuss key observations regarding:
 - Accuracy and performance on the non-linear dataset.
 - Decision boundary shapes and smoothness.
 - Sensitivity to hyperparameters (e.g., kernel width for KDE, k for KNN).
 - Computational complexity and convergence behavior (if applicable).

Deliverables: students need to deliver the following

1. Video Recording (25–30 minutes):

- A comprehensive walkthrough of the code and methodology.
- Explanation of the mathematical equations implemented for each estimator and classifier.
- Demonstration of the code execution, showcasing the decision boundary plots for each task.
- Final interpretation and comparison of the results from Task 6.

2. Python Notebook (IPYNB):

- Well-structured and commented notebook containing all the code.
- Clear markdown sections for each task, explaining the approach and observations.

Critical Notes:

- **Implementation Restriction:** Use of high-level libraries like scikit-learn for building the estimators (e.g., *GaussianMixture*, *KernelDensity*, *KNeighborsClassifier*) or classifiers is **strictly prohibited**. Students must implement all algorithms using the equations and mathematical formulations discussed in class.
- **Permitted Libraries:**
 - Only general-purpose libraries are allowed: *numpy*, *matplotlib*, *pandas*, and *scipy* or *sklearn* (for basic functions like *multivariate_normal.pdf*).

Deadline: 20th July 2026